



OysteRestoration

An *Ostrea edulis* restoration project in North Adriatic sea

CALOCERO ANDREA

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Part 1: **The species** —

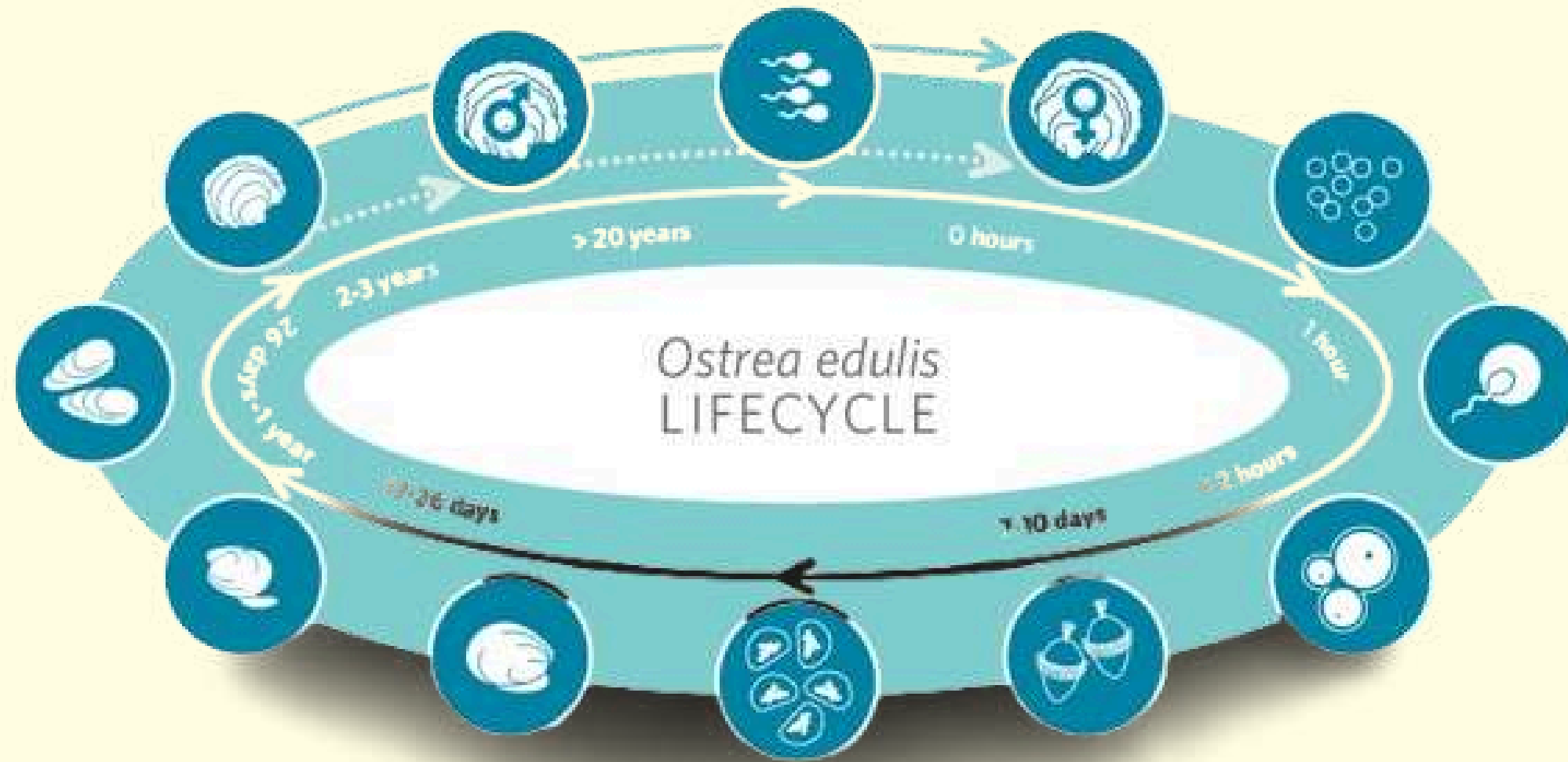


Ostrea edulis, also known as european flat oyster or native oyster, is a bivalve belonging to the ostreidae family, typically found along the coasts of the Atlantic Ocean, the North Sea and the Adriatic Sea

They prefer **rocky** substrates, although they can also be found on **muddy** and **sandy** ones as well, where they form **tridimensional structures**, know as beds or reefs, that provide many ecosystem services



Biological cycle



O. edulis is a **protandrus alternating hermaphrodite** species that starts as a male once mature (3 years), and turns female after the first spawning event

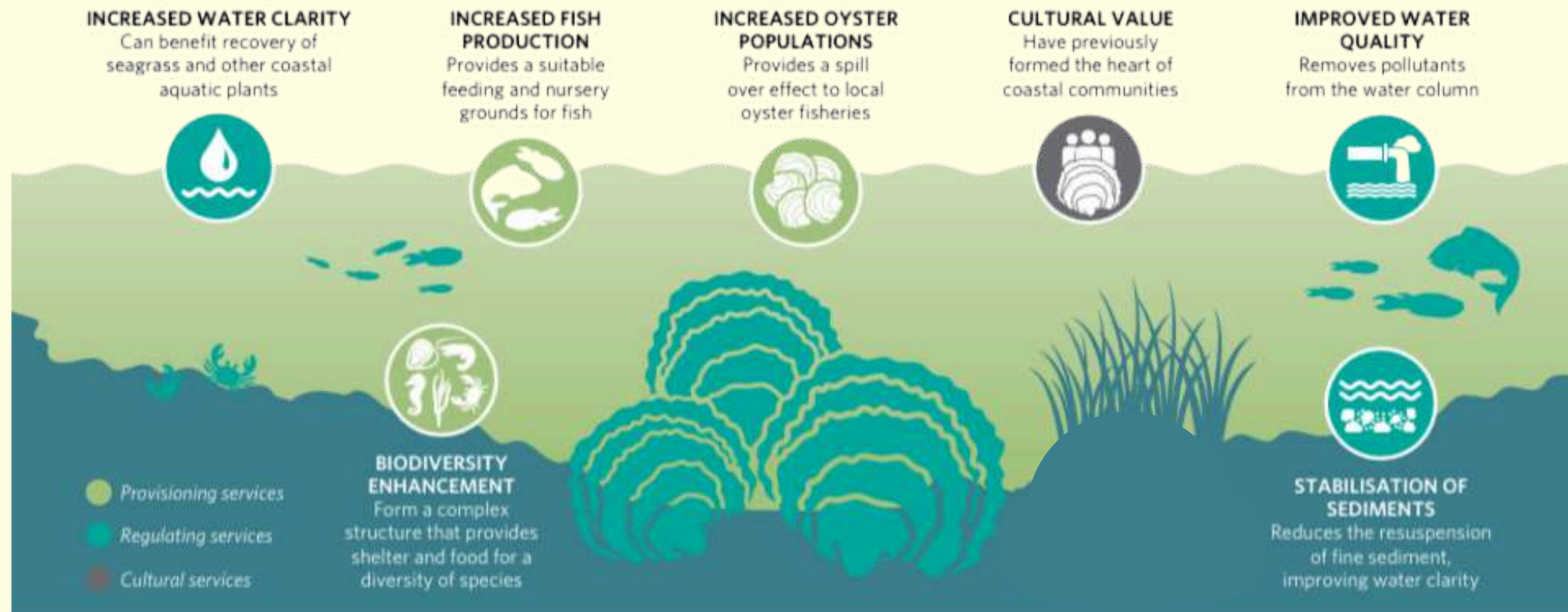
After the first sex change, individuals will begin to **alternate** between male and female **repeatedly** in the same reproductive season

The **reproduction period** spans from **June** to **September** .

Usually at higher temperatures we have a sex ratio unbalanced towards males, so in warmer months reproduction will be less effective

Ecological role

O. edulis is considered an **ecosystem engineer** for its filter feeding strategy and ability to create complex substrates that provides numerous ecosystem services



Unfortunately in N-Adriatic these services are **extremely reduced**, given the depletion and scattering of residual oyster populations in the area

Health status and decline

Natural oyster beds, especially in N-Adriatic are **strongly declining** due to several factors, among which **overharvesting** played a key role, being unregulated for the entire 19th and part of the 20th century

IMPACT ON POPULATION

- Reduced broodstock
- Reduced larval input and survival
- Reduced larval settlement and survival

FACTORS AFFECTING NATIVE OYSTER LIFE STAGES

Anthropogenic

Biotic and abiotic

Anthropogenic-biotic

Overfishing



Water quality and pollution



Habitat loss



Capital or maintenance dredging



Environmental conditions



Predation



Skewed sex ratio



Pest species



Pathogens



Invasive species



Today only **few small residual beds** can still be found in **N-Adriatic**, threatened especially by bottom trawlers, beam trawlers fishing and habitat loss

Part 2: The project —

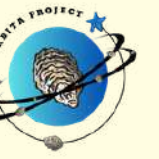


O.R.B.I.T.A.

Oyster reef Restoration Back Into The Adriatic sea

The scope of this project is to restore a portion of the once abundant *O. edulis* beds in the Adriatic sea, bringing back these important ecosystems with all the benefits they provide

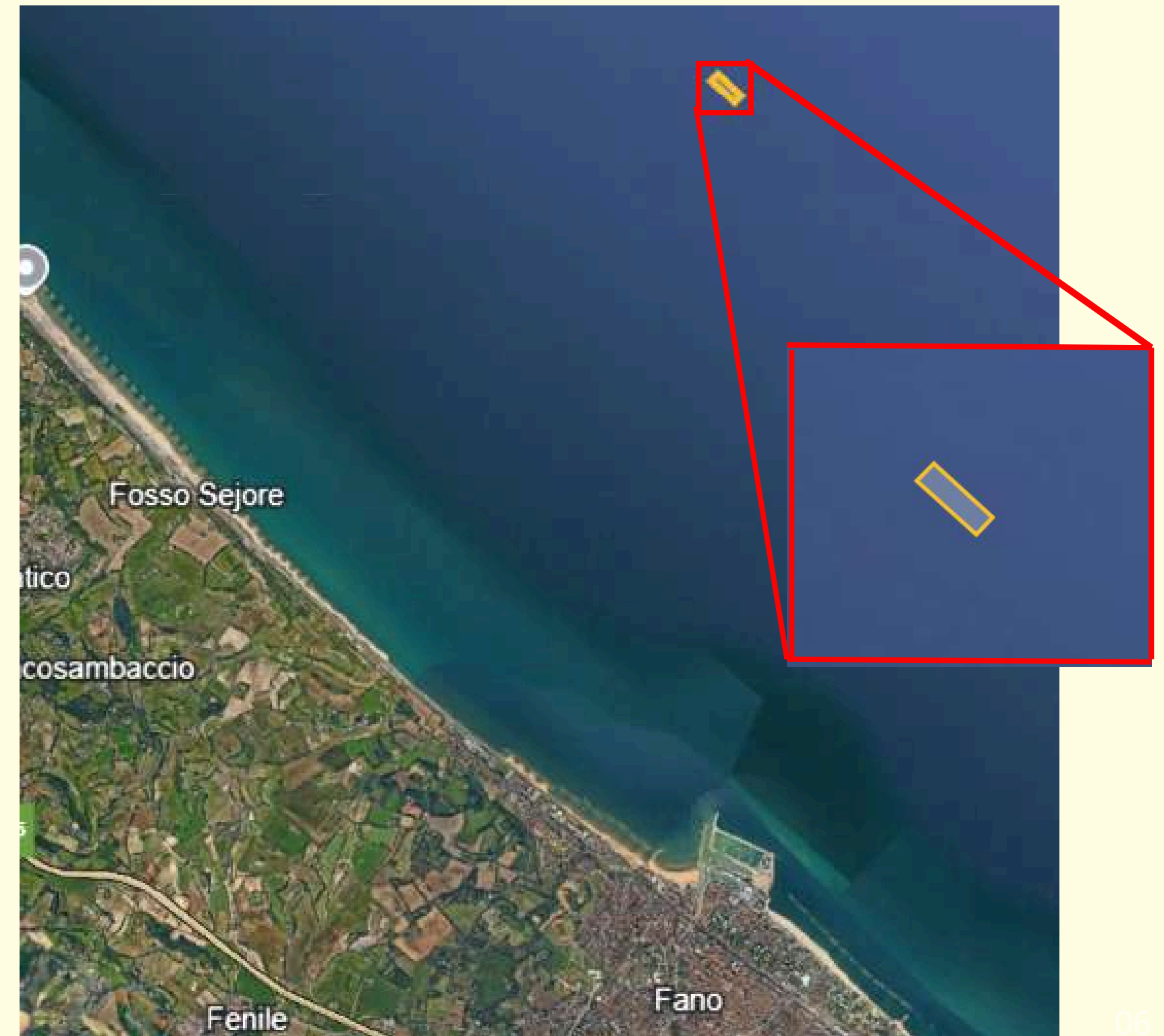
Site selection



For our restoration project we chose a **2 hectares** area, in front of the **Fano coast**, around **3.5 Nm** from the coast, at **10 m depth**, near to some of the actual oysters' occurrences

The site selection occurred following a careful analysis of:

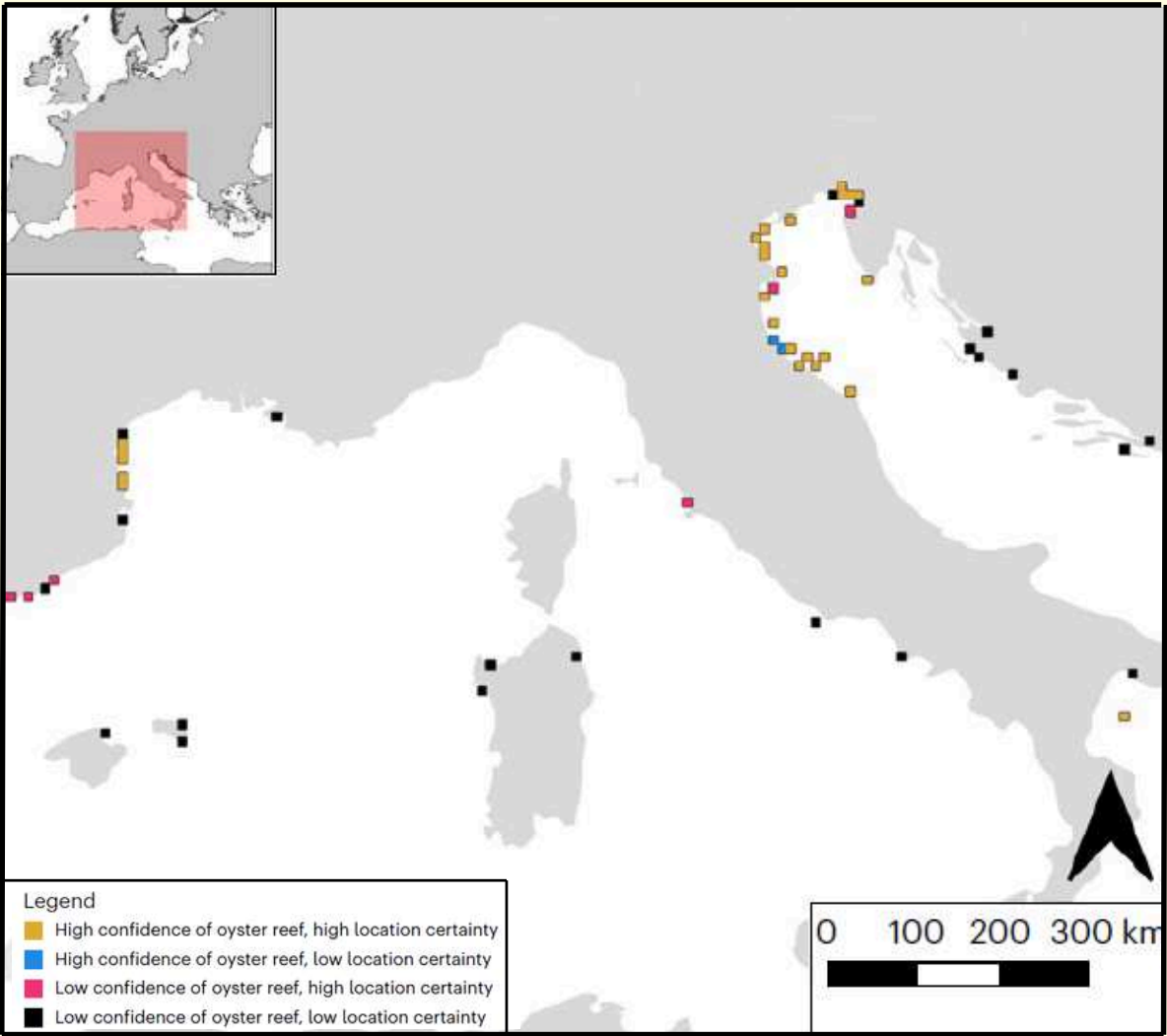
- **Historical records**, regarding ancient oyster reef distributions in the Adriatic sea
- **Actual distributions**, obtained from the OBIS and GBIF occurrences databases
- Information obtained through **interviews** with **Marche region fishermen**



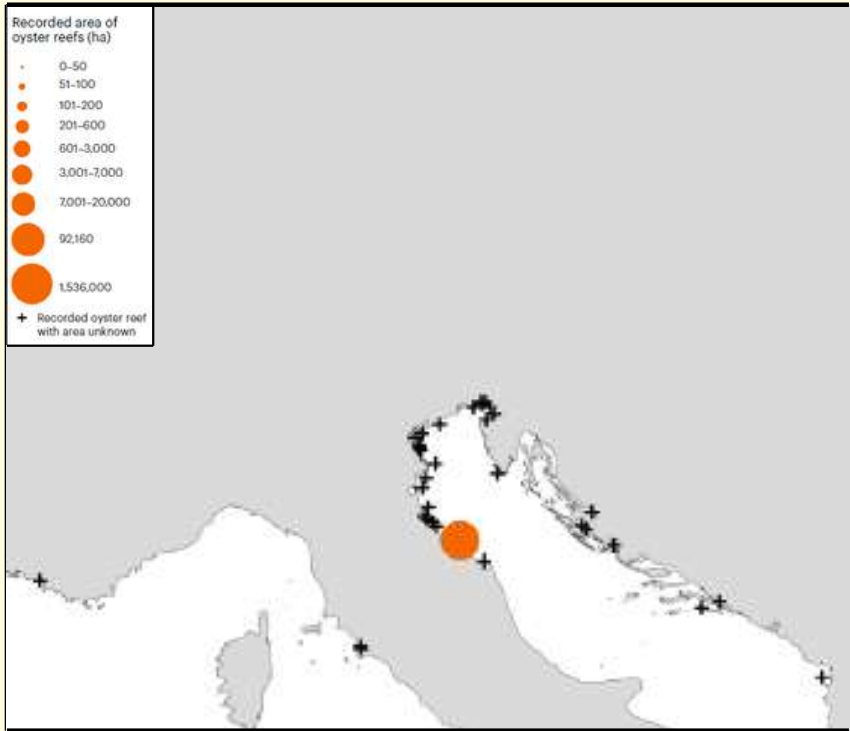
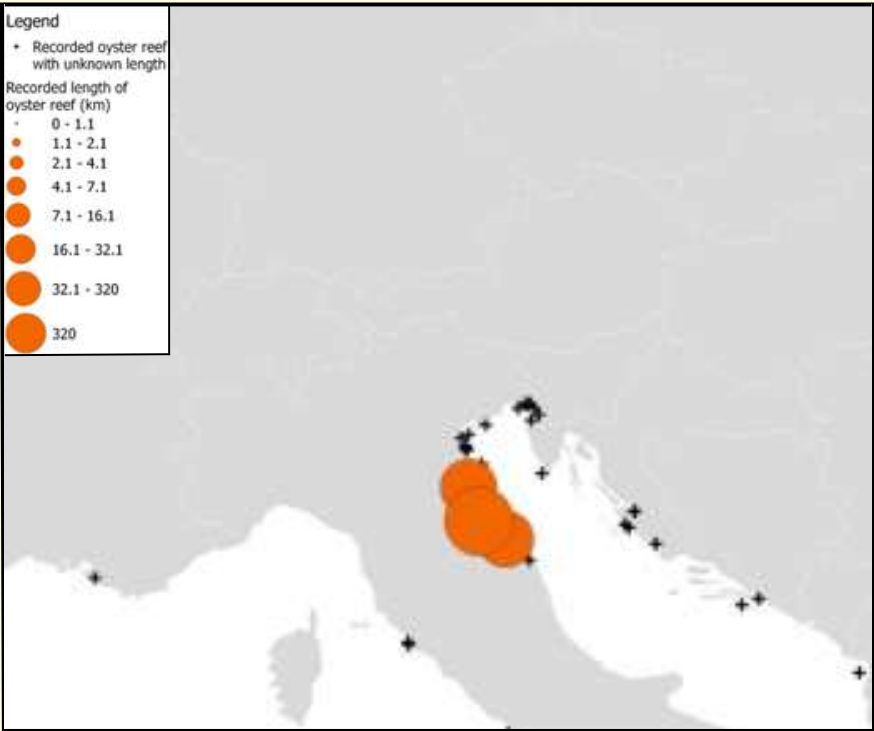
Historical Records

Basing on the scientific paper: *Thurstan, Ruth H., et al. "Records reveal the vast historical extent of European oyster reef ecosystems." Nature Sustainability (2024): 1-11*; in which all historical evidences and data, regarding the ancient distribution of *O.edulis* in Europe, were collected (using many sources); We considered its ancient distribution in the **Adriatic Sea**:

We noticed that **in the past these reefs were quite widespread** in this area, with high location certainty



The only reef in the **Adriatic sea** of which we have **historical informations** regarding **length and extent** was located in the **Northern Marche region, between Fano and Pesaro**



"Oysters are found mostly on rocky shores in 2-5 fathoms; on a bank to the south-west of Grado, near the estuary of the river Isonzo; on a smaller bank west of Izola, near Capo d'Istria; near Pula and Novigrad, east of Zadar; along the coast of San Cassano [Sukošan], and on the Scogli Ostia and Galisniac [islets Oštarije and Galešnjak]. They occur also near Sebenico, Stagno [Šibenik, Mali Ston], & c. [...], near Brindisi. Ancona. Punto di Maestra and Chioggia. and near the mouths of the rivers Po, Adige, and Brenta." (1883^[88])

"There is another bank in front of Fano and far from the beach four miles. In this place, it is twelve steps deep and four hundred long and extends towards the northwest as far as Pesaro. It begins again in Rimini continuing up to Cesenatico [...], and then starts opposite Primaro again, ending above Magnavacca." (1715^[89])

[88] Faber, G. L. The Fisheries of the Adriatic and the Fish Thereof. A report of the Austro-Hungarian Sea-fisheries (B. Quaritch, 1883).

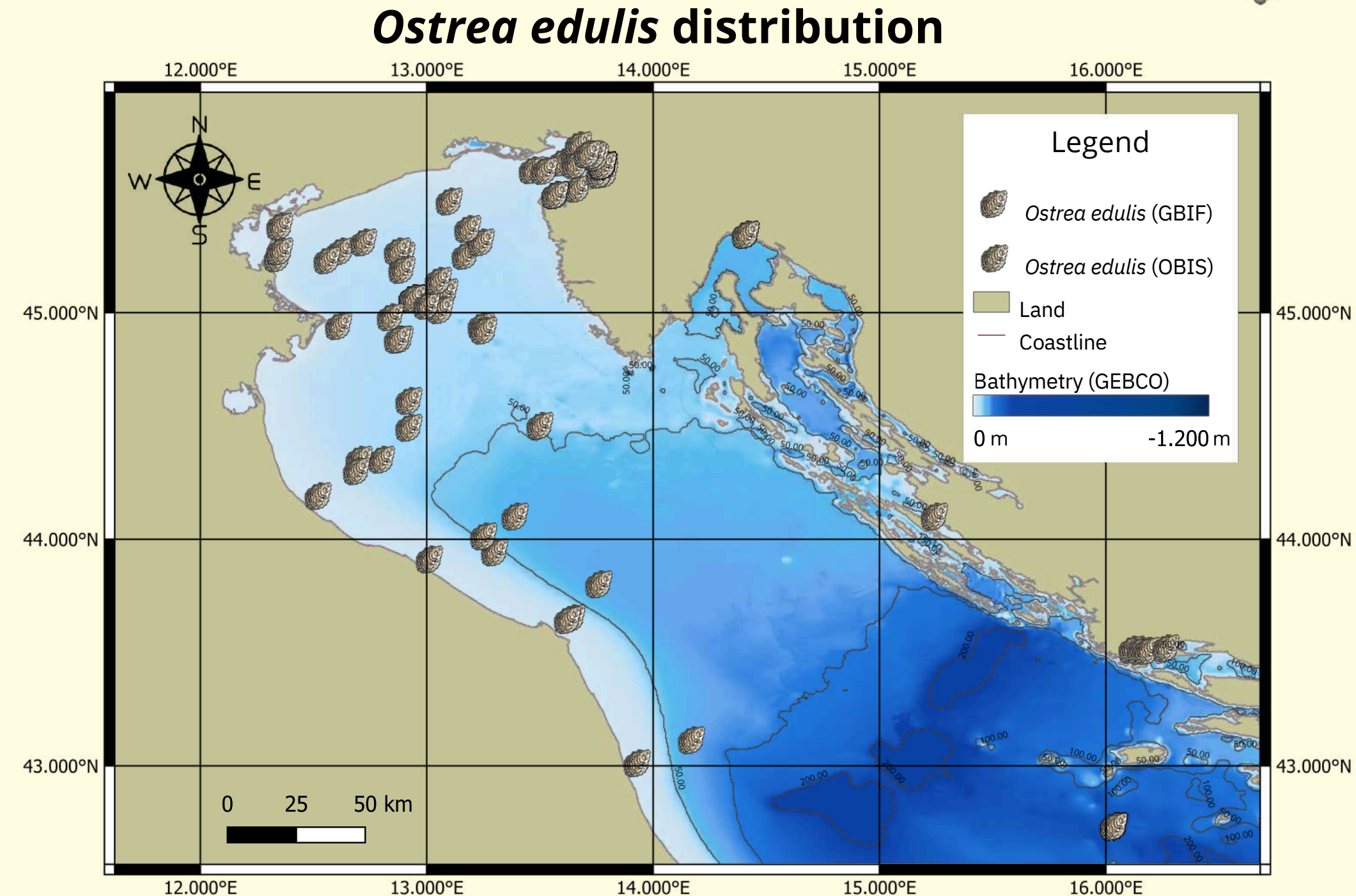
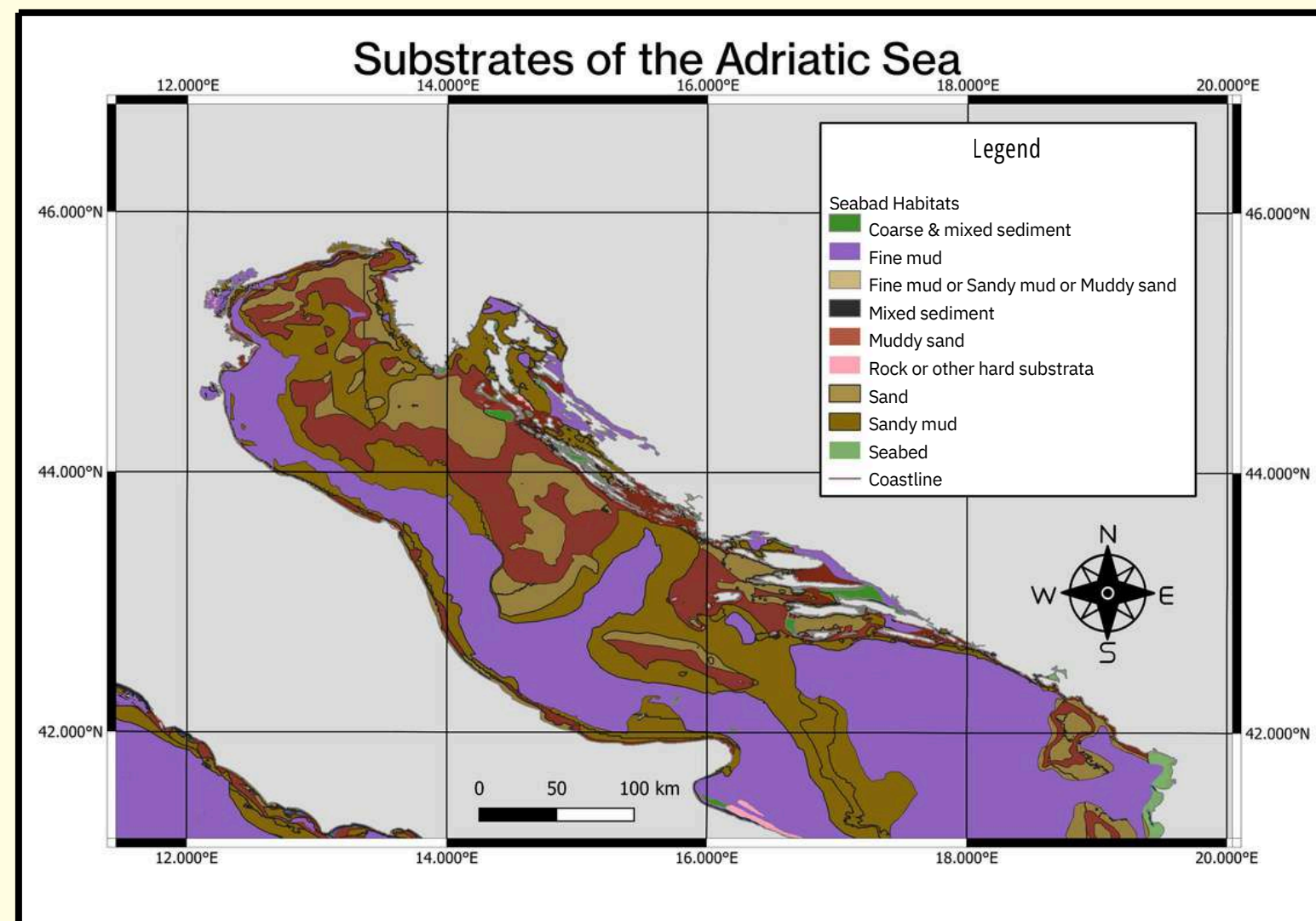
[89] Marsili, F. Giornali e memorie per la ricognizione delle spiagge pontificie dell'Adriatico (Biblioteca Univ. di Bologna, 1715).

Actual distribution

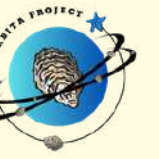


We studied the **actual *Ostrea edulis* distribution** in the Adriatic sea, exporting the occurrences data on **GIS maps**, in order to have clear in mind the actual situation, noticing how, nowadays, oysters are **mainly distributed** in a **patchy way** than in reef structures (as in the past).

We took in mind also the **substrates distribution** and the **currents patterns** in the area



Local fishermen interviews



We went to the **Ancona's port** for several mornings, with the goal of interviewing some local fishermen in order to get some **information on the past and current oysters' status in the Marche region**. Indeed, we believe that **L.E.K.** is crucial in the site selection for a restoration project.

The questions we asked the fishermen were:

- Which are your usual fishing areas?
- Do you ever catch oysters?
- Based on your knowledge, are there oysters in the northern-central Adriatic?
- If so where?
- Are them more or less abundant than in the past?



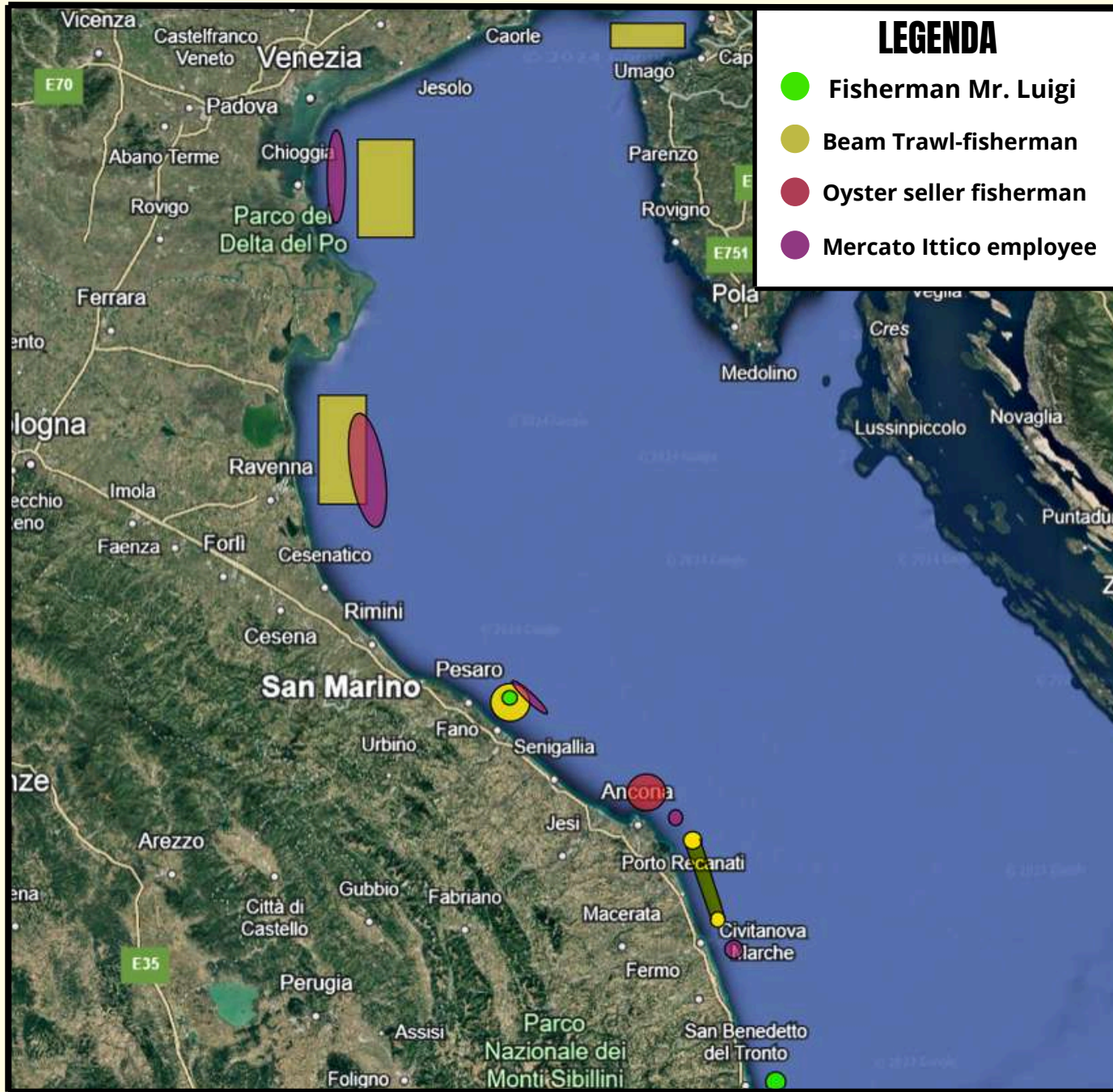
We collected their answers, trying to reconstruct a **oysters' distribution map, based on the L.E.K.**

Young Beam Trawl-fisherman

"I fish with beam trawl between the **Conero area** and **Civitanova Marche** and it can happen to catch some oyster here but probably, in the **Northern part** of the **Adriatic sea** are more abundant while in the **Northern part of Marche region** there a few more oysters than here, but not that much, in fact oysters are not our fishing target but, when we catch them, we are happy the same"

"Since I started working as fisherman, the number of Oysters' chatches in the area I fish is more or less the same"

LEK DISTRIBUTION MAP



Mr. Luigi, Fisherman from Ancona

"It the past 15-20 years there were several oysters' aggregations in the **Ancona's area** but nowadays there are very few left, from some fishermen friends I know that there are a little bit more **near Fano** and **San Benedetto del Tronto**"

"I think that their disapparence is mainly due to the loss of their habitats, especially with changes in the seabed structure"

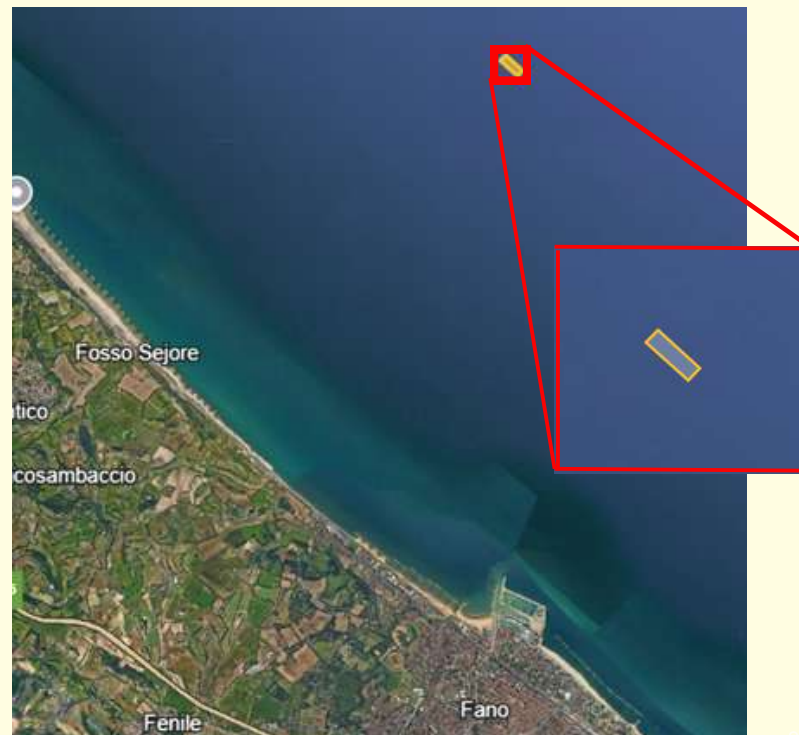
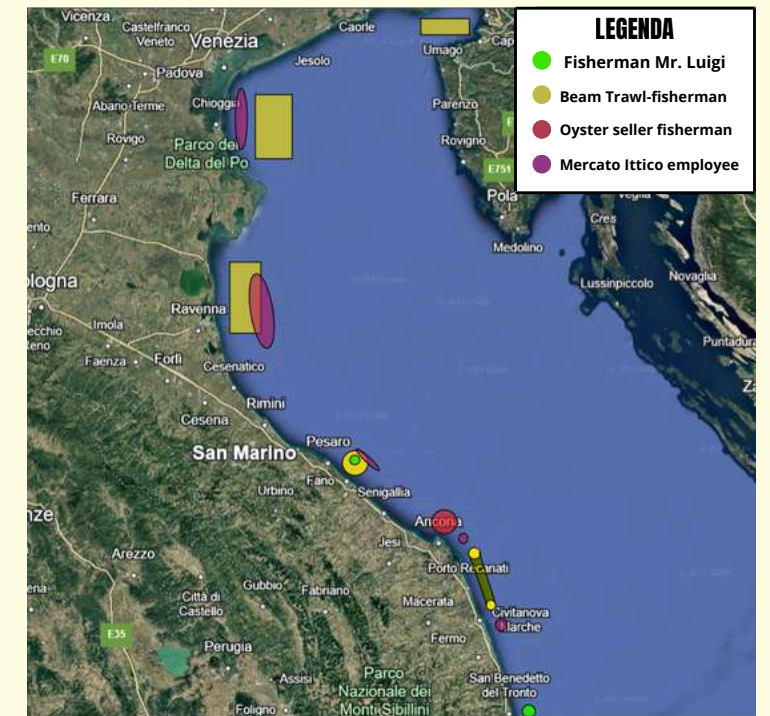
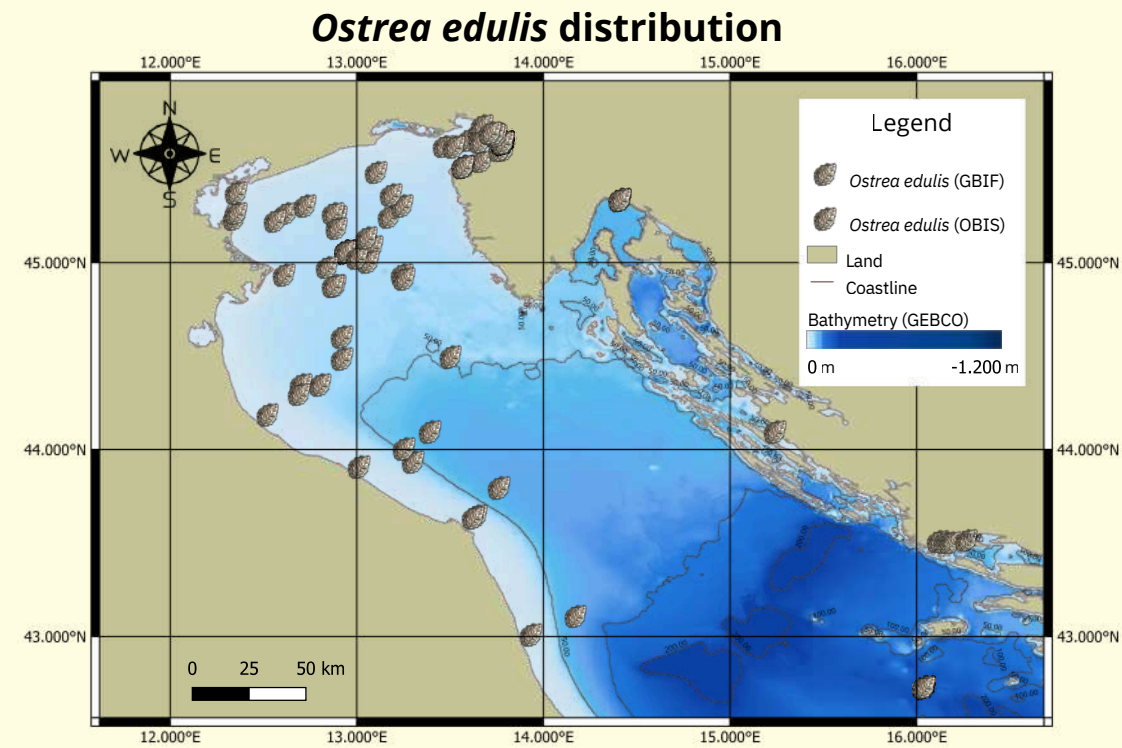
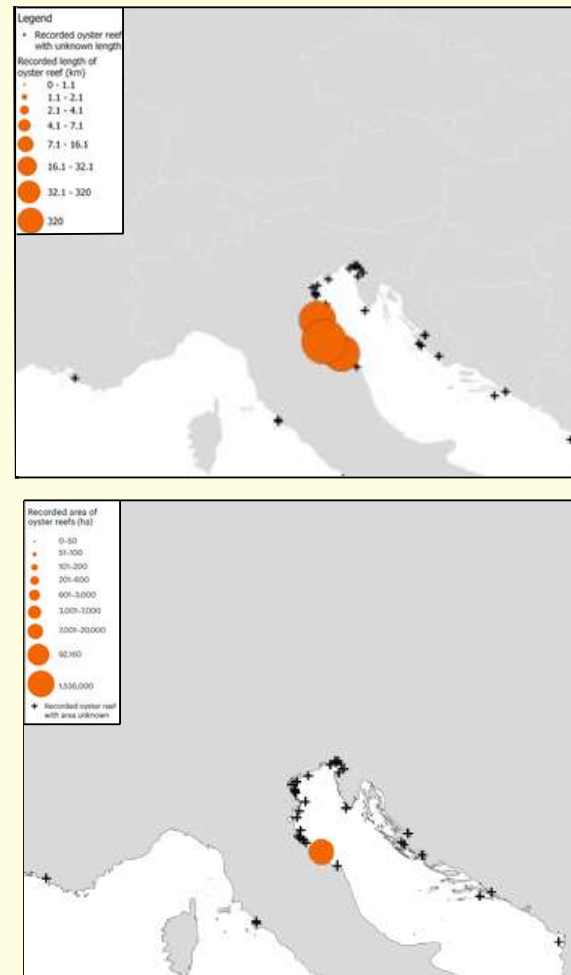
Ancona's fish market employee

"Here at the fish market, we have oysters very rarely as there are not many in the area. Until a few years ago a larger number of them could be found but currently they are a rarity, a few more are caught going further **north of Ancona** and sometimes fishermen catch some, now and then, **near Civitanova**; in the **Northern Adriatic** there are more oysters than here".

Fisherman selling oysters at the Ancona's port

"There are some oysters here in the **Ancona's nearby**, mainly around 3.5 Nm from the coast but they are not that common and for sure they're less than in the past, especially here".

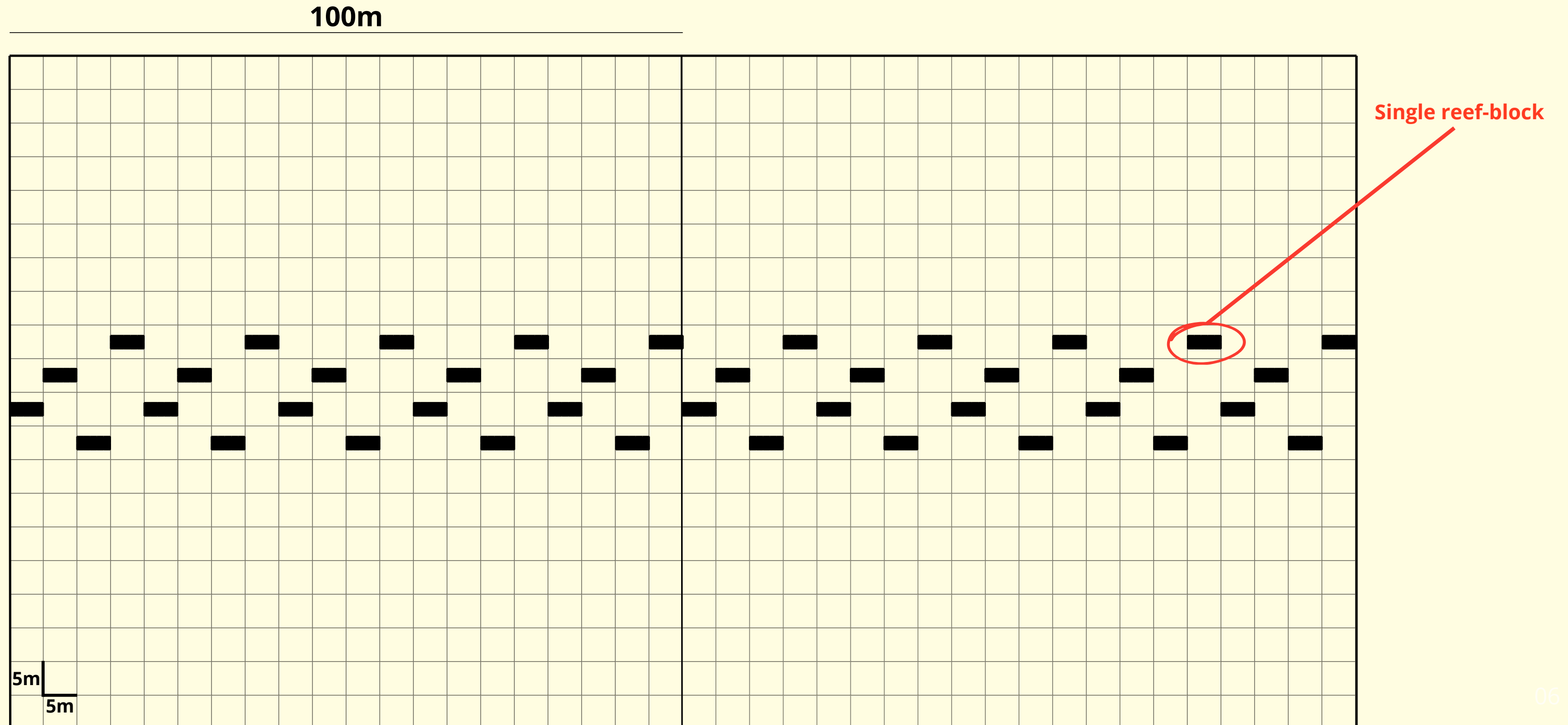
Cross-referencing all the obtained data, we concluded that the Fano-area could be an optimal restoration site

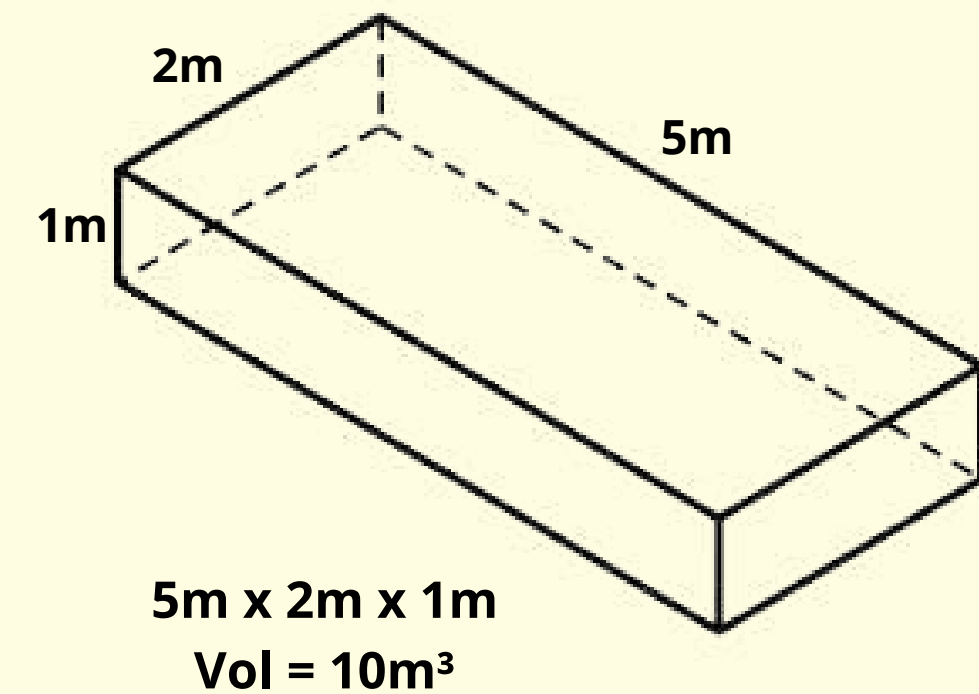


Restoration Design



In our restoration plan we want to place **40 reef-blocks**, inside the 2 hectares of selected area, in **4 rows**: each row is **3m** apart, with the blocks in the single rows being **15m** away from each other. The selected area is placed downstream from the residual natural oyster aggregation to take advantage of **natural larval dispersal**





Each reef-block (**10m³**) is composed by a **metal cage**, filled for one half with **inert calcareous stones** and the other half with recycled **oyster shells**. We chose this materials because they are both surfaces on which oysters' larvae have optimal settlement rates

Why parallelepiped structures ?

- Generating **3D structures**, **increasing** the **complexity** of the substrate
- **Reducing** the **sinking** of seeded oysters into the muddy substrate, **decreasing mortality**, which would be high by seeding them directly on the seafloor
- Offering 4 surfaces available for larval settlement on the 4 lateral faces, plus the face on which the mature oysters are seeded, providing a **large settlement surface**
- Immediate **coastal protection from hydrodynamism**

The single reef-structures are deployed in their final location on the seafloor by using a **vessel** equipped with a **crane on a mechanical arm**

System used in the Life NatuReef, Ravenna

We estimated that the deployment of all the reef-blocks will take 4-5 days

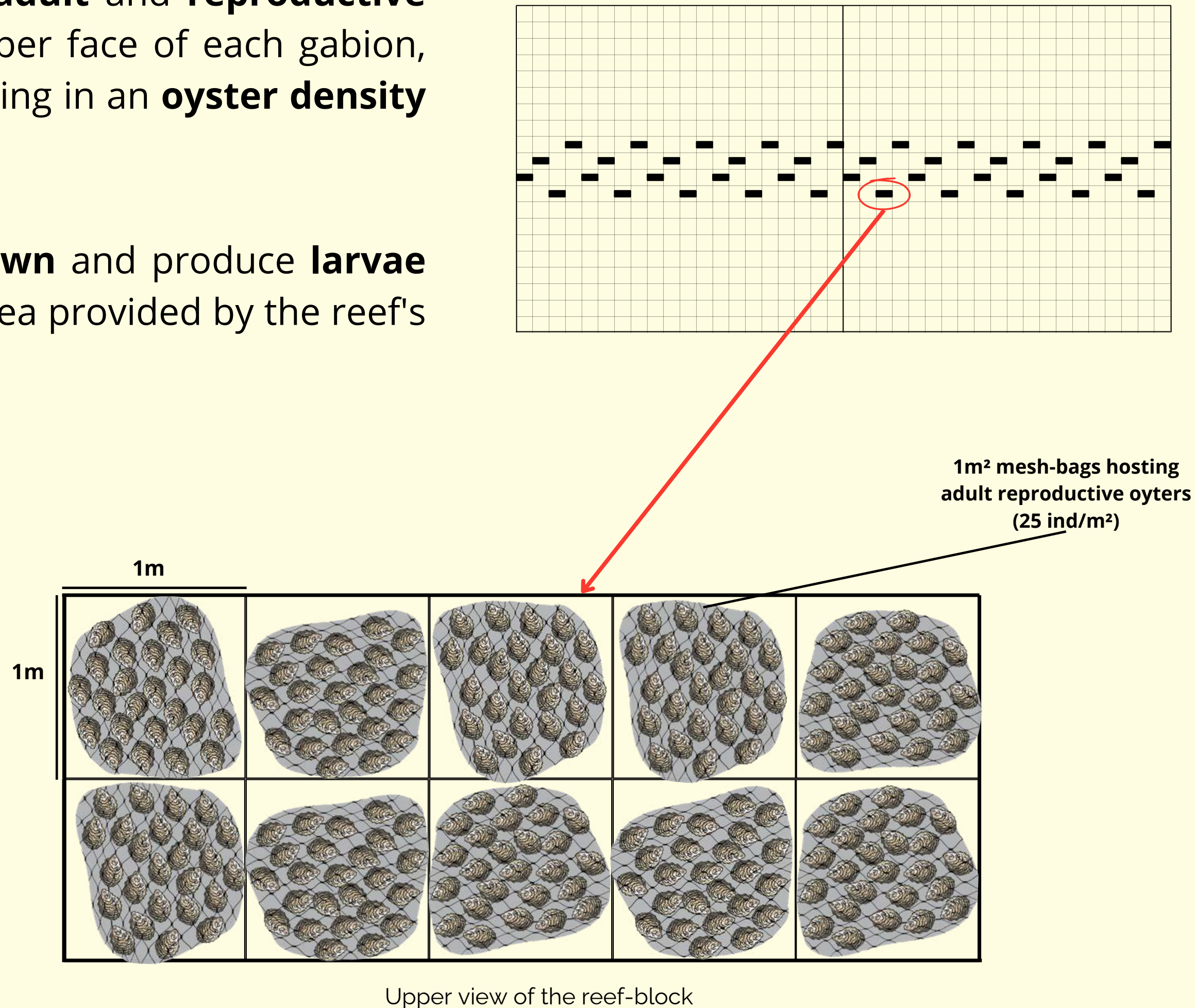


10 mesh-bags, 1m x 1m, containing **25 adult** and **reproductive** individuals each, are attached to the upper face of each gabion, before placing them on the seabed, resulting in an **oyster density of 25 ind/m²**.

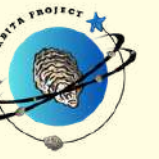
These mature oysters will be able to **spawn** and produce **larvae** that can **settle** on the available surface-area provided by the reef's 3D structures.

In this way, on the 40 reef-blocks we'll **seed 10,000 mature oysters**, which will serve as **broodstock** for the future reef expansion and to sustain the population

The presence of adult alive oysters **enhances** the **larval settlement** rate and **oyster shells** are proved to be the **favourite substrate** for larvae to settle on



Biosecurity and Legal Considerations



Adult Oysters

Adult oysters are purchased from Naturedulis hatchery (Goro), with a **health and biosecurity certification**.

Health certification is also crucial for oyster trading and transportation operations, as it helps prevent the spreading of disease to new areas.

Oyster Shells

Step 1: Washing shells with a **high-pressure water jet**

Step 2: Shells are **sterilized** by UV sunlight by **weathering** for **6 months** to inactivate pathogens and parasites, **turning them every 2 months**

A **1 hectare field will be rented**, out of the city, to perform this activities

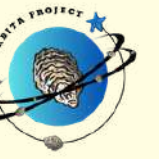


System used in the Shuck don't chuck project; Australia

All the activities involved in the restoration project must follow **international standards, specific regulations** and **legal frameworks**

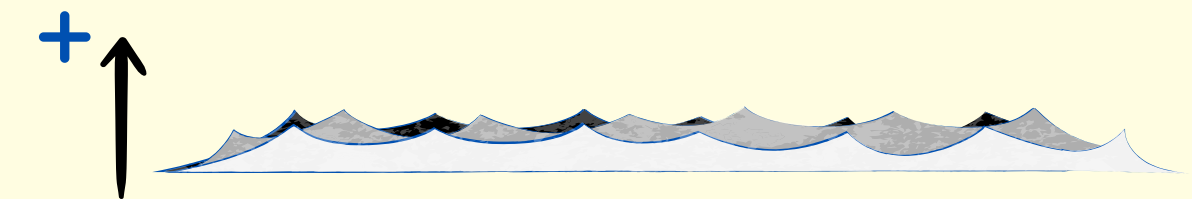
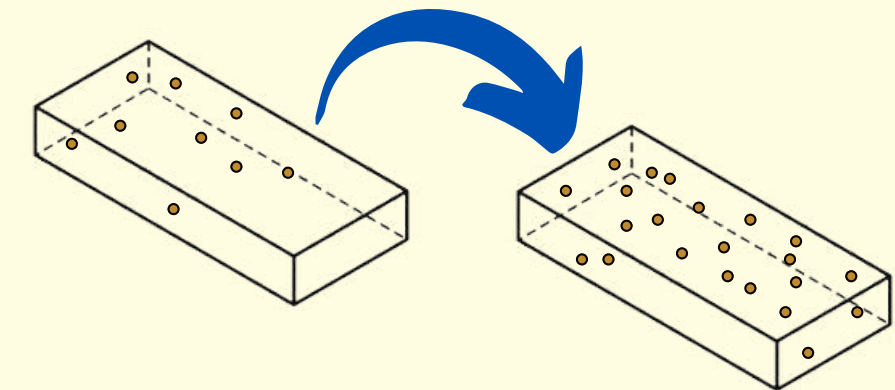
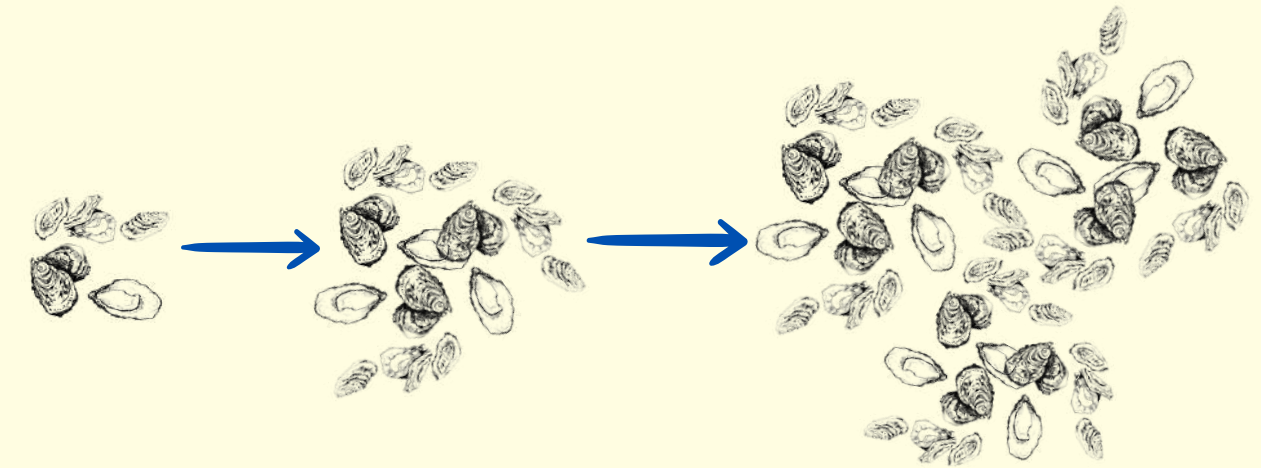


Success conditions

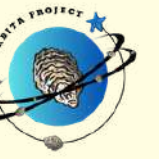


We set **4 goals** to define whether this project can be considered a **success**:

- Positioned broodstock shows an **high survival rate** on the long term, with oysters in **good health, growing** and **expanding the bed**
- We observe an **increase in larval recruitment** on blocks over time
- We observe an overall **improvement of water quality** in the area
- We observe an **increase in biodiversity** associated to the artificial bed



Project Monitoring



The monitoring will be conducted on the **natural oyster bed** and the **restoration site** considering specific metrics and evaluating at the same time some **environmental parameters**. We will follow a BACI design

Natural bed

- Habitat progression
- Oyster density
- Cover percentage
- Sex ratio

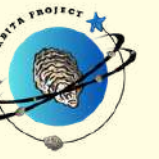
Restoration site

- Growth rate
- Survival rate
- Sex ratio
- Recruitment rate

Environmental parameters

- Water quality
- Biodiversity
- Temperature
- Salinity

Monitoring: **Natural bed**



Habitat progression

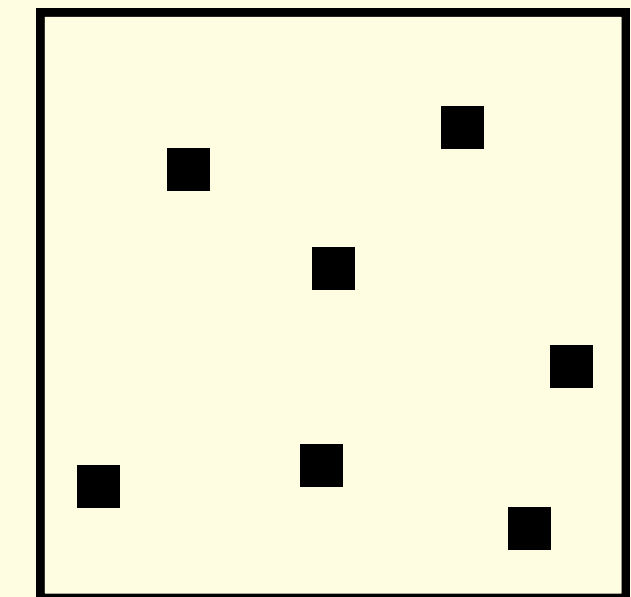
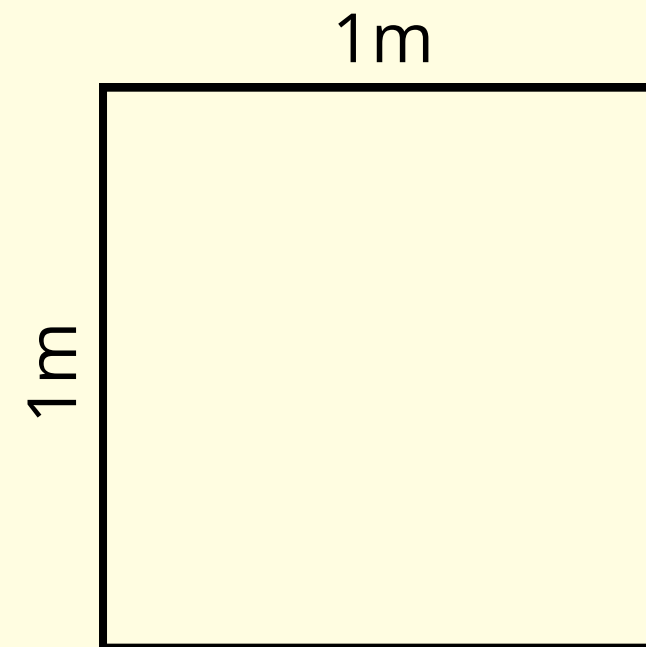
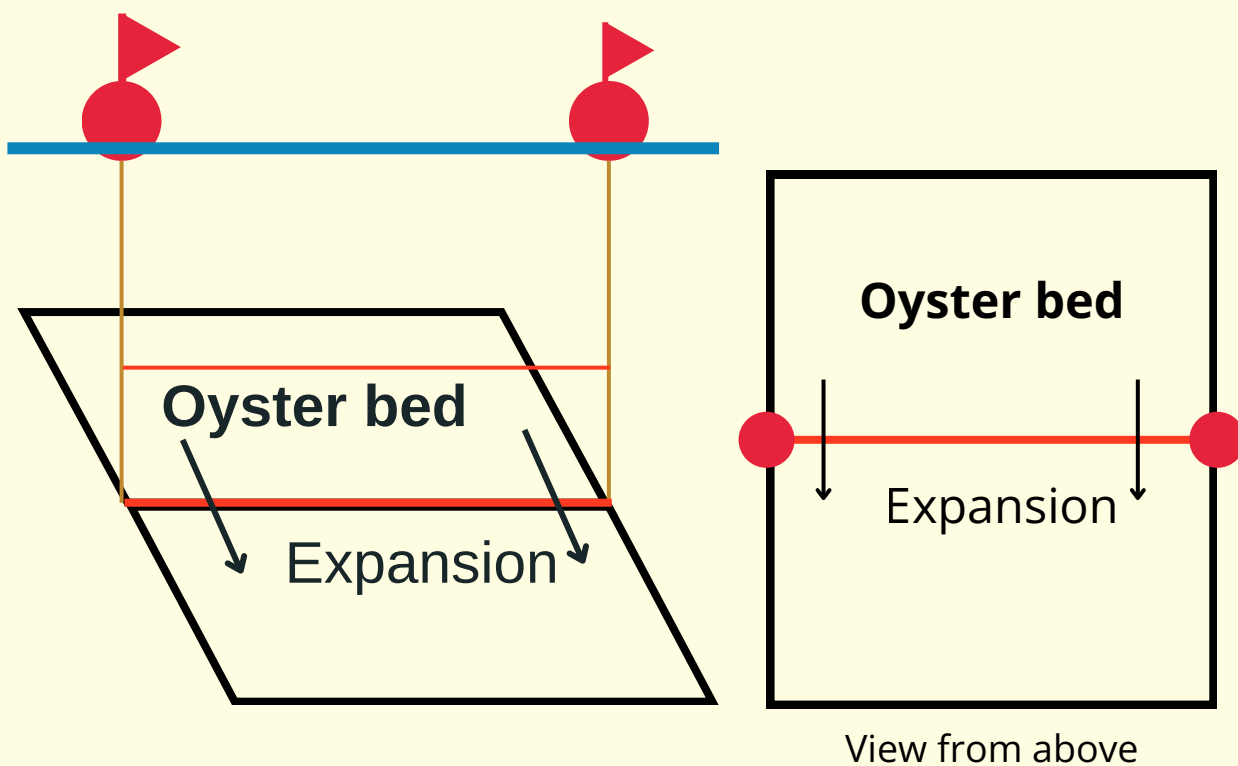
Two buoys tied together to a line suspended near the substrate as a reference point

Oyster density and Cover percentage

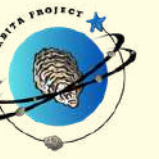
1m x 1m photo quadrats

Sex ratio

Random sampling

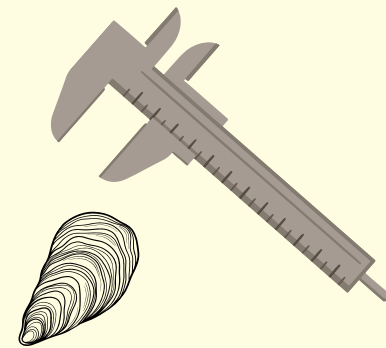


Monitoring: **Restoration site**



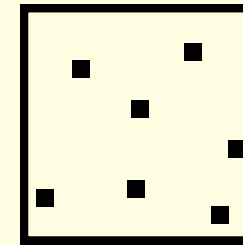
Growth rate

Visual census coupled with photographs and random measures with a caliper



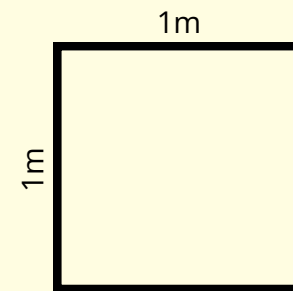
Sex ratio

Random sampling and gonad evaluation of individuals



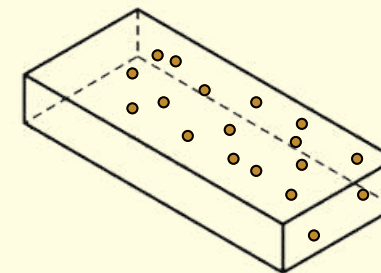
Survival rate

Visual census of randomly chosen blocks

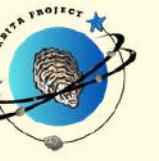


Recruitment rate

Counting of new recruits per unit area

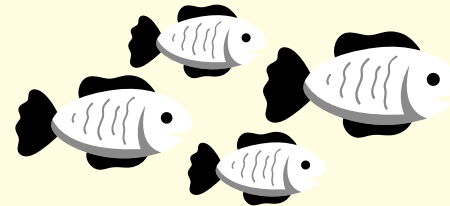
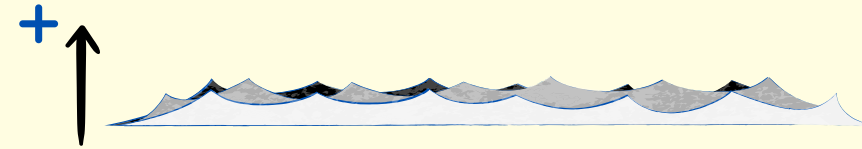


Monitoring: **Environmental parameters**



Water quality

Water sampling with Niskin bottle



Biodiversity

BRUV (Baited Remote Underwater Video) system for vagile and swimming organisms
Photographs and ImageJ analysis for sessile organisms

Salinity

In situ measurements with CTD (Conductivity Temperature Depth) sensors

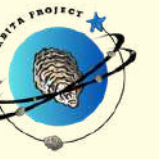
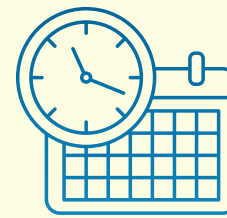
NaCl



Temperature

In situ measurements with CTD sensors

Time schedule



Natural bed

11.5 years of analyses, in this way we acquire a baseline

4 samplings each year, once every 3 months

Start **January 2026-**
End **September 2037**

Restoration site

10 years of analyses, starting after the blocks have been positioned

4 samplings each year, once every 3 months

Start **September 2027-**
End **September 2037**

Environmental parameters will be assessed **continuously** since **January 2026**

Timeline

Preliminary work

- Shell collection
- Broodstock production in hatchery
- Natural bed monitoring
- Environmental status assessment

**January 2026-
August 2027**

August-September 2027

Restoration actions

- Preliminary gravel deposit
- Positioning of blocks with broodstock
- Natural bed monitoring
- Environmental status assessment

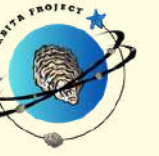
Long-term monitoring

- Oyster monitoring
- Natural bed monitoring
- Environmental status assessment

**September 2027-
September 2037**

The monitoring will be carried on for 10 years to appreciate reef development

Part 3: Stakeholders —



Involving as many people as possible in the project is crucial to its success, **Informing** people about the restoration work is an essential element of developing networks, awareness and impact of any project, helping in the funding requirement, gathering more donors and volunteers

There are **multiple parties** which can **potentially** be **affected** by the **restoration** operation

It is important **to collaborate**, working together towards a **common goal** that can benefit all the involved parties

To **collect** the huge amount of **oyster shells**, needed for restoration (**200 m³**), it is essential to **involve fishermen, restaurant owners and the population**:

Fishermen

Usually oysters are considered as **discard** and the shells must be disposed as **special waste**, with associated costs. We would take those shells **for free** from fishermen, **recycling** them for the restoration project.

At the same time it is important to **let fishermen understand** that this restoration action, in the future, could enhance their catches and should not be considered as a threat to their work



Restaurant owners



As oyster shells are just **useless** for **restaurants**, our aim is to establish a **partnership with local restaurants**, taking their waste-shells, which would otherwise be thrown away, **recycling** them

Partner restaurants will receive the “**oyster friendly**” **stamp** and the **list of partner restaurants** will also be available on the **project website**

Oyster-friendly restaurant

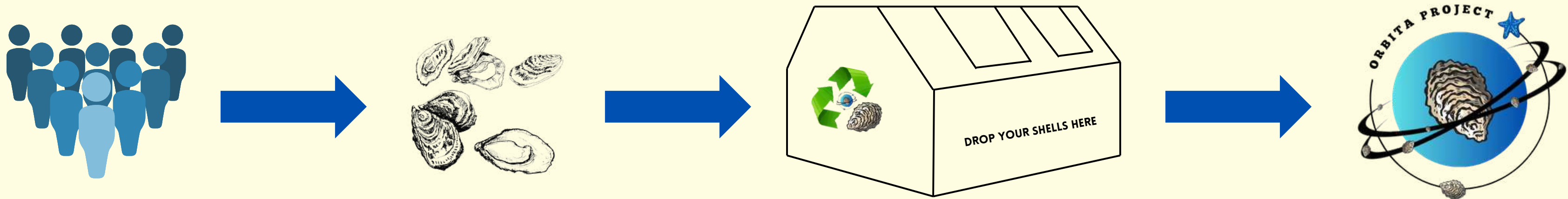


Local population

It is important to **involve** the **local population**, thereby increasing the sense of ownership of the project, increasing potential volunteers and donors and reducing potential opponents

The population can be involved in the **recycling of shells** from domestic use by placing “**public drop points**” in inhabited places near the restoration area

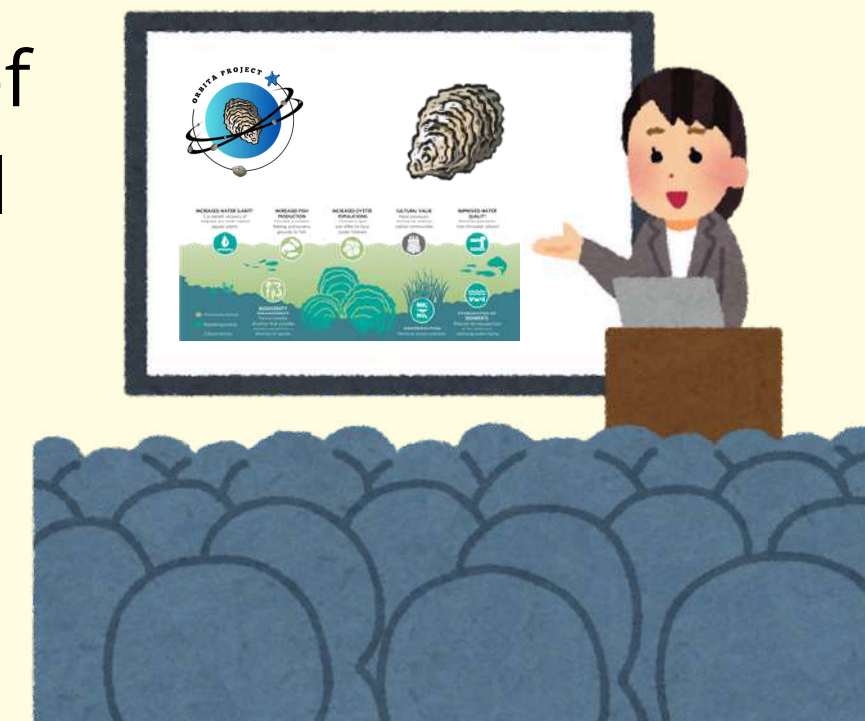
As used in the Oyster Recovery Partnership project in the Chesapeake Bay



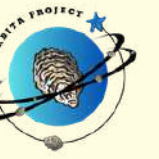
Raising awareness and **informing the population** about the importance of biodiversity and the ecosystem services provided by oyster reefs is also crucial

A **3 seminars-cycle** will be held by scientists from different universities:

1. **Biology and ecosystem services provided by oyster reefs**
2. **Main threats to oyster reefs**
3. **Our Restoration Plan: the ORBITA PROJECT**

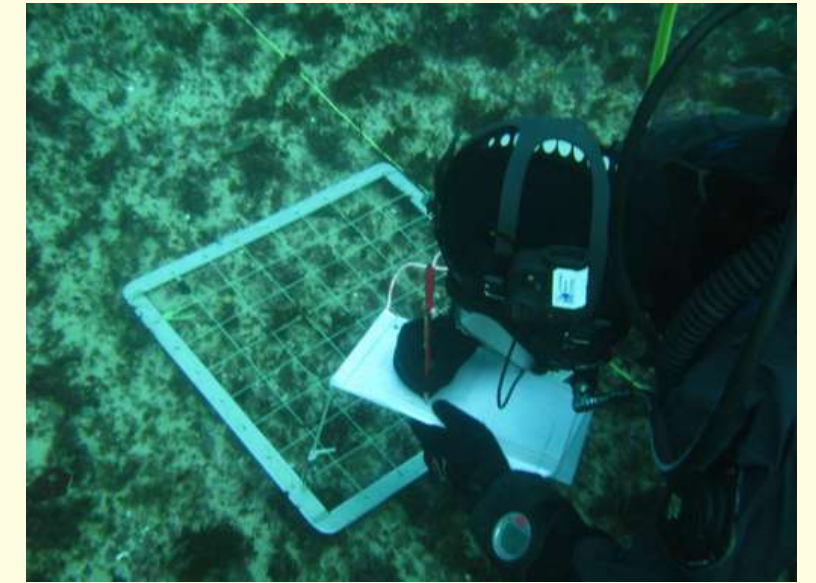


Diving centers



Can be involved in 2 parts of the project:

- In the **short term** they can help in the monitoring with **visual census**, through **citizen science programmes**
- In the **long term**, once the increase in biodiversity associated with the reef has occurred, the site can be exploited for **recreational oyster reef-dives**

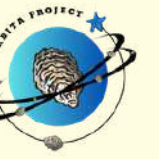


A **percentage of the revenue** from divers in this area can be agreed with the diving centers, to be **donated to the project**

Universities

UNIVPM and **UNIBO** (already involved in the NatuReef project) : **Marine Biology students** could be involved in different phases of the project, mainly in monitoring activities

Professors from the two universities could participate as guests in public seminars and conferences



Associations

Volunteering associations as **Fano UniMar** can be commissioned to hold **workshops** about the importance of oyster reefs and can **coordinate citizen science activities**

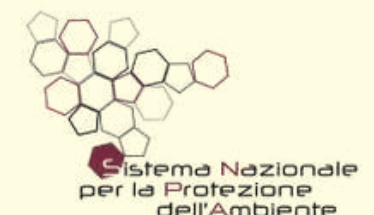
Reef Ceck Italia can be involved too, organizing training courses for citizen science



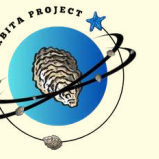
Authorities



Research institutes



Communication



Communication is essential to **let people care** and contribute to the project

It's important to have **high visibility** on the main **social media**, letting people know what we are doing, with periodical updates, enhancing the chances for economical support

For this reason a **strong social-promotion** is needed



Nowadays is vital having a **website** containing all the infos about the project, news, events and details on how to get involved



**Oyster reef Restoration Back
Into The Adriatic sea**

[Our Work](#)

[Get Involved](#)

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There will be the possibility to **donate** for the project or to **buy** our **merchandise**, in order to **sustain our work**

In both cases the donor will receive by e-mail an “**Oyster Friend certificate**” and a **brochure** about **oyster reefs ecology** and **threats** and notions about the **ecosystem services** provided by them, underlining the importance of **scientific dissemination**



Part 4: Cost Analysis —

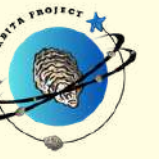


The **distribution of costs is uneven over time**, with costs being much **higher at the beginning** with reef-block construction and positioning being more **resource intensive** while monitoring activities are more costly but diluted over time

Restoration	Broodstock: 70.000€	Calcareous rocks: 15.000€; Gravel: 9.000€	Cages: 25.000€	Boat with crane + crew 33.000€	Blocks construction 10.000€	Biosecurity: 6.000€	Oyster shells: In kind contribution + 60€/m³ of incentive for restaurants (max 12.000€, considering the 200m³ of shells) Oyster drop points: 5.000€	Total 185.000€
Monitoring	Equipment: 4000€	Research divers: 170.000€	Rent boat: 50.000€	Lab analysis equipment + salaried personnel: 15.000 €				239.000€
Communication	Seminars: 6.000€	Social promotion: 10.000€	Workshops: 10.000€					26.000€

TOT: 450.000 €

Founding opportunities



European

Interreg Euro-Med

LIFE
PROJECT

EU Horizon
Programm

MedReAct

National

CNR-National
Biodiversity Future
Center
(NBFC)

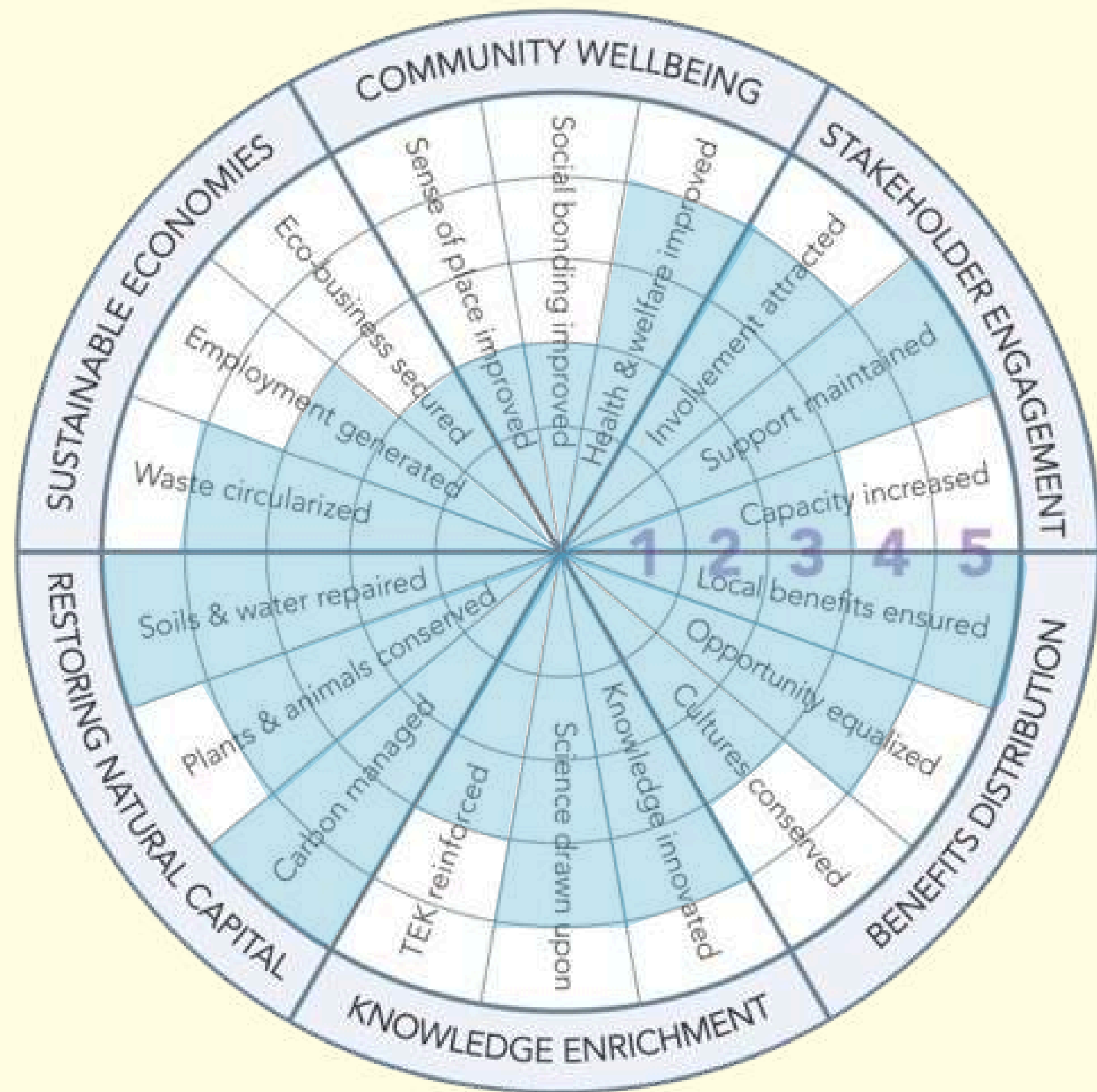
Private

Private investors

Donations

Merchandise

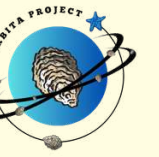
Social benefits in the future



To assess the **progression** of our restoration project we decided to use the **social benefits wheel**, that includes a **1-5 stars system**

This is the social benefits projection after the 10 years long-term monitoring

Expected Ecosystem services coming from our Restoration



SUPPORTING

- **Biodiversity enhancement:** providing shelter and food for several species, increasing complexity
- **Increase water clarity,** removing resuspended material with **sediment stabilization**

PROVISIONING

- **Increased fish production:** providing foraging and nursery grounds
- **Oyster spill-over effect** to local fisheries, in the long term

REGULATING

- **Coastline protection:** from waves energy and reducing coastal erosion
- **Carbon sequestration:** acting as Carbon sink

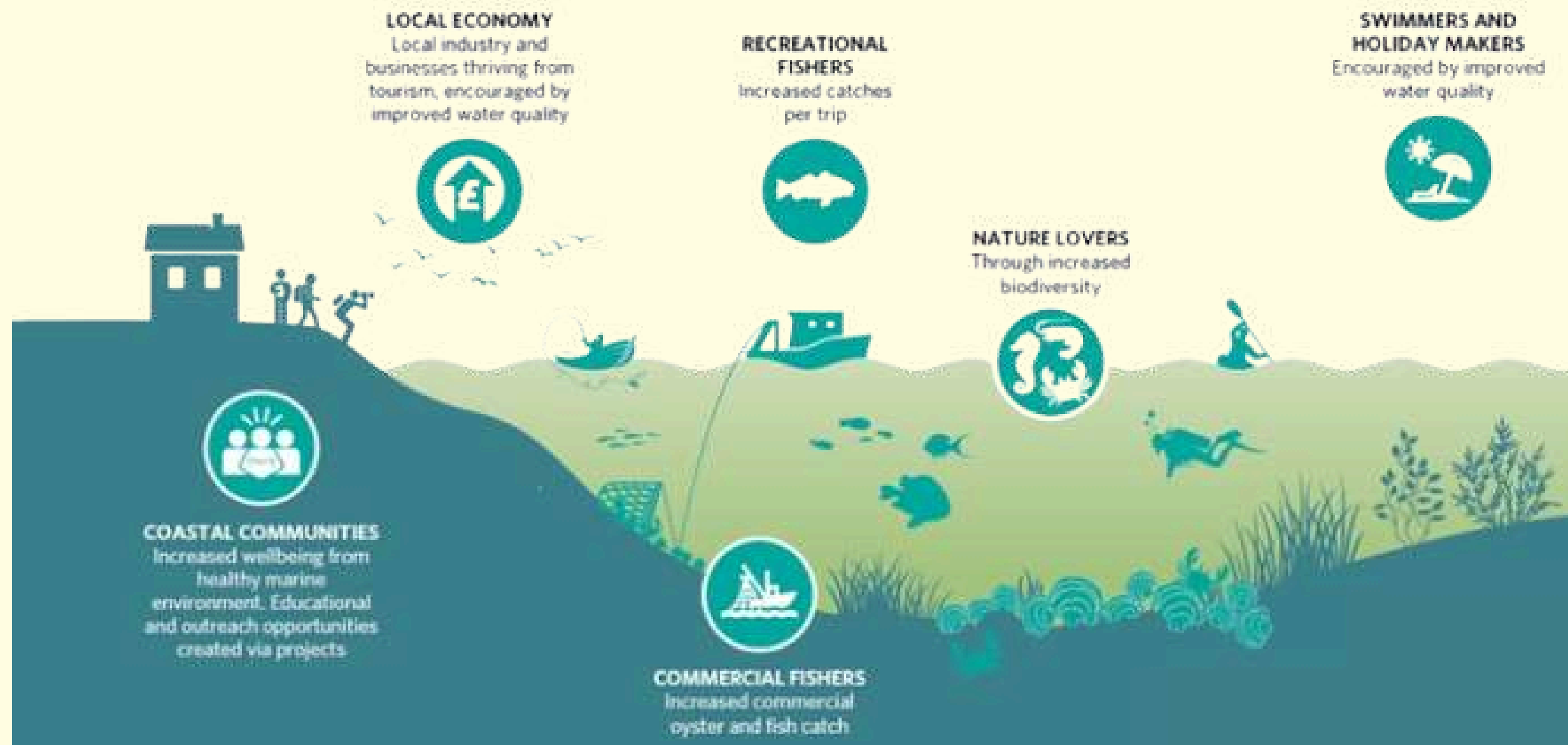
CULTURAL

- **Educational opportunities** for the population about the importance of biodiversity
- **Recreational opportunities:** Oyster reef dive and snorkeling tourism, in the long term
- **Increased wellbeing** of coastal communities

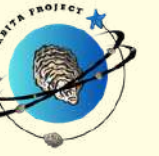


Beneficiaries of Oyster Restoration

This restoration plan will be **beneficial for several parties**



Return On Investment

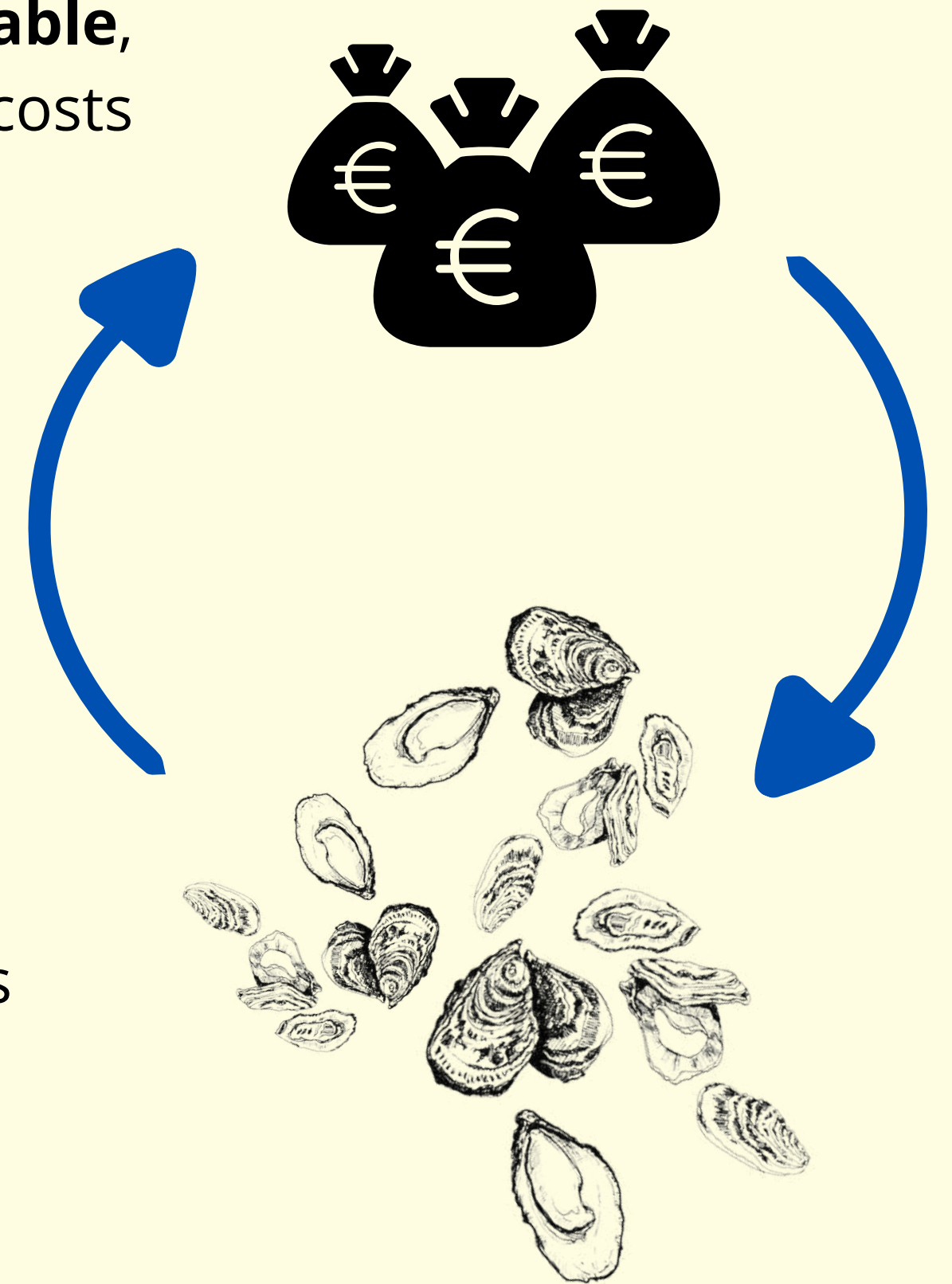


Generally, the ROI for an oyster restoration project is quite **variable**, ranging from **2 to 14 years** to completely recover the restoration costs in benefits and ecosystem services.

This variability is due to many factors such as techniques, substrate, choice of proper site etc.

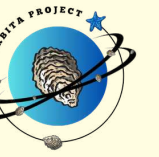
Size, however, seems to have a major influence as 75% of projects working on areas **>0.4 ha show a positive ROI**.

Given this information, we expect a **complete return** of the costs **within 10 years** since the blocks' positioning.



Ecosystem Service	Value (€/ha*year)
Recreational fishing	900 €
Commercial fishing and harvesting	15.000 €
Submerged vegetation enhancement	1.400 €
Bacterial removal and Carbon burial	Undefined
Shoreline protection	45.000 €
Oyster reef ecosystem natural value	up to 90.000 €

SWOT Analysis



Strengths

Biodiversity and ecological value
Economic and social benefits
Community involvement
Institutional support
Research

Weaknesses

High initial costs
Continuous monitoring costs
Slow recovery of populations
Vulnerability to disease



Opportunities

Ecological and educative tourism
Science research and innovation
Technological Innovation
International Collaborations

Threats

Climate Change
Pollution and contamination
Anthropogenic pressures
Economic and political pressures

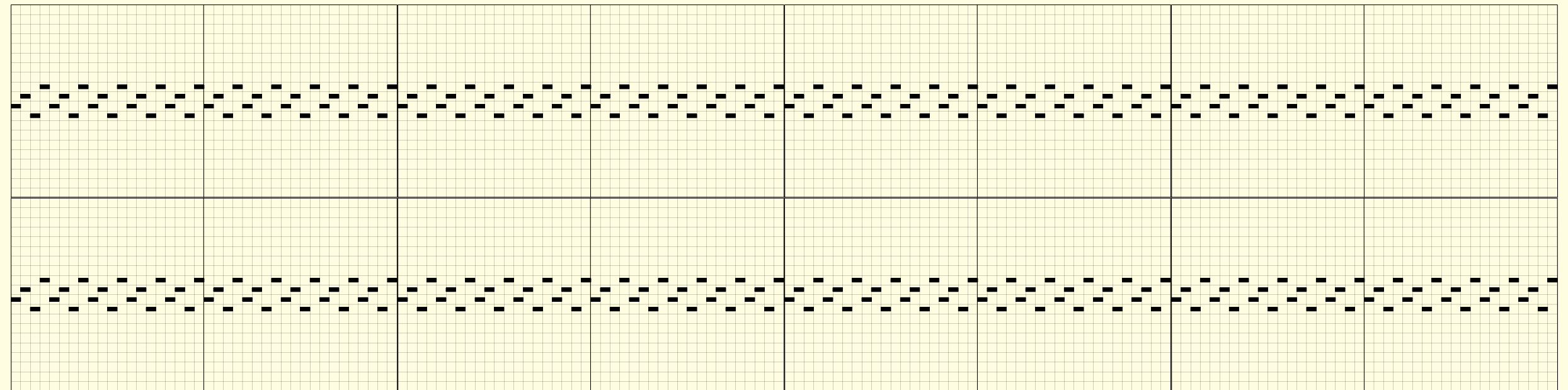
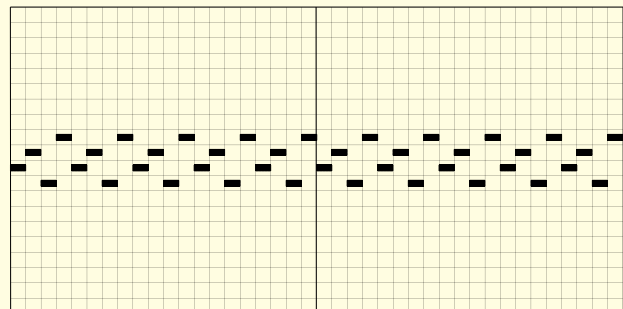


Future perspectives

This restoration project is a **small-scale project**, considering “just” **2 hectares** of restoration area

Our aim is to demonstrate “**in small**” the **effectiveness** of our restoration plan, in order to obtain the **funding necessary** for a project, covering a **wider restoration area**

INITIAL PLAN



FUTURE (?)



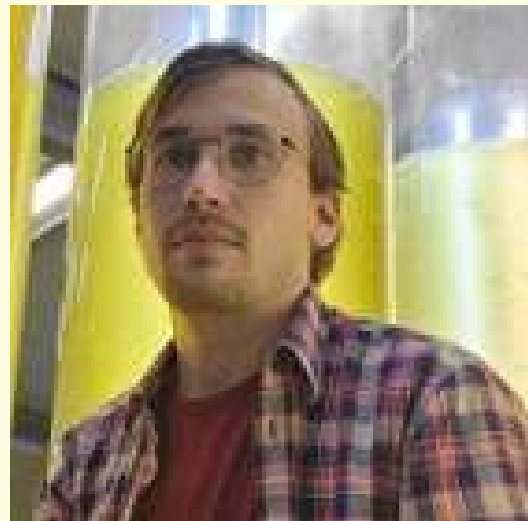
A special acknowledgement to the experts who provided us useful advices for our final project



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References

- Bolton, Adam, and Anita Nedosyko. "Windara shellfish reef construction." Australasian Coasts and Ports 2019 Conference: Future directions from 40 [degrees] S and beyond, Hobart, 10-13 September 2019: Future directions from 40 [degrees] S and beyond, Hobart, 10-13 September 2019. Hobart: Engineers Australia, 2019.
- Branigan, Simon, James Fitzsimons e Chris L. Gillies. Middens moderni: riciclaggio delle conchiglie per il ripristino di un ecosistema marino in via di estinzione a Victoria, in Australia. Vol. 21. N. 3. 2020.
- Grabowski, Jonathan H., et al. "Economic valuation of ecosystem services provided by oyster reefs." Bioscience 62.10 (2012): 900-909.
- Harding, Simon, Lauren Nelson, and Tim Glover. "Solent oyster restoration project management plan." Blue Marine Foundation (2016): 47.
- <https://marinesciences.uconn.edu/mstc/vesselops/rates/>
- <https://omnicalculator.com/>
- Preston, Joanne, et al. "European native oyster habitat restoration handbook." Zoological Society of London, 2020.
- Robertshaw, Gareth, Anita Nedosyko, and Atilla Bayram. "Windara Reef-lessons learnt from the first large scale oyster reef restoration in Australia." Australasian Coasts and Ports 2019 Conference: Future directions from 40 [degrees] S and beyond, Hobart, 10-13 September 2019: Future directions from 40 [degrees] S and beyond, Hobart, 10-13 September 2019. Hobart: Engineers Australia, 2019.
- Thurstan, Ruth H., et al. "Records reveal the vast historical extent of European oyster reef ecosystems." Nature Sustainability (2024): 1-11.
- Zu Ermgassen, Philine, et al. "European guidelines on biosecurity in native oyster restoration." Zoological Society of London, 2020.
- Zu Ermgassen, Philine, et al. "European native oyster habitat restoration monitoring handbook." (2021).

- Robertshaw, Gareth, Anita Nedosyko, and Atilla Bayram. "Windara Reef-lessons learnt from the first large scale oyster reef restoration in Australia." Australasian Coasts and Ports 2019 Conference: Future directions from 40 [degrees] S and beyond, Hobart, 10-13 September 2019: Future directions from 40 [degrees] S and beyond, Hobart, 10-13 September 2019. Hobart: Engineers Australia, 2019.
- Thurstan, Ruth H., et al. "Records reveal the vast historical extent of European oyster reef ecosystems." Nature Sustainability (2024): 1-11.
- Zu Ermgassen, Philine, et al. "European guidelines on biosecurity in native oyster restoration." Zoological Society of London, 2020.
- Zu Ermgassen, Philine, et al. "European native oyster habitat restoration monitoring handbook." (2021).

**THANKS FOR
YOUR
ATTENTION !**

